Russia national innovation strategy

One page country assessment | Team 2 | Steve, Gloria, Tiger | 10 November 2026

"Russia's national innovation system combines substantial technical capability with constraints on financing and institutional performance. In WIPO's fixed 2025 comparison, Russia ranks 60th overall, versus the United States at 3rd, China at 10th, and France at 13th. Russia's human-capital-and-research pillar ranks 28th, while institutions rank 131st. These indicators suggest that useful technical inputs coexist with weaknesses elsewhere in the innovation system; they do not establish that education alone determines national performance. [R5-R8; AI-1]"

"Education is therefore a resource to sustain and deploy. Factbook-derived entries report Russian literacy at 99.9% (2021), school life expectancy at 15 years (2023), and education expenditure at 4.2% of GDP (2023). Spending shares for the United States, China, and France are 5.4% (2021), 4.0% (2023), and 5.3% (2022). The mixed years and input measures cannot rank classroom quality or research productivity. Relative to the United States' university advantage and China's population scale in the game, Russia should seek the most productive affordable improvement; France's cheaper secondary-education upgrades make imitation of its spending mix unattractive. [R1-R4; G1; AI-1]"

"Technology strategy should connect existing knowledge to realizable output. The Factbook-derived Russian profile describes a large space program, illustrating concentrated technical capabilities. In the game this legacy is represented by +7 starting tacit knowledge, compared with France's +4; Russia also starts with the lowest wealth. Real-world GDP measures must remain separate: the same archive puts 2024 Russian GDP at $2.174 trillion at official exchange rates but $6.089 trillion in purchasing-power terms, above France's $3.732 trillion PPP figure. The instructor's normalized wealth, not a GDP calculation of our own, controls the game budget. Russia should prioritize attainable applied research and fund basic research when it unlocks a usable path or supplies valuable tacit knowledge. [R1, R4; G1; AI-1]"

"Intelligence and immigration require a balance between access and retention. The game gives Russia closed immigration, a defensive bonus, and vulnerability to talent outflow; the United States and France begin open, and China begins closed. The Russian archive's positive aggregate net migration estimate, 0.76 per thousand in 2025, cannot establish whether researchers are arriving or leaving. We therefore should not infer skilled migration from population totals. Keep the starting posture provisionally, test the actual talent-flow formula, and open only if the expected human-capital benefit exceeds the loss of protection. Compare affordable espionage opportunities with applied research and saving instead of treating espionage as assured catch-up. [R1; G1; AI-1]"

"Proposed decision rule: seek agreement, adopt a two-of-three majority, and let Steve resolve an unresolved split or time-limit deadlock after hearing both czars. All three members participate in executive review and recordkeeping. Our recommended strategy uses public basic-research spillovers without assuming they supply private tacit knowledge, maintains productive research from the opening round, and preserves some wealth because savings enter the final score. The plan seeks improvement from Russia's baseline as well as absolute technology points. This rule and strategy remain proposals until the team meets. [G1; S1, p. 4; A1; AI-1]"

AI generated review draft. Quoted italic text requires independent student review. Citations, exact prompt, and factual critique follow in the appendices. No team consensus is asserted.

Education recommendation for Tiger

Team 2 Russia | Proposed portfolio text for independent review | 10 November 2026

"I recommend selective education investment that improves the effectiveness of research without exhausting Russia's limited budget. The real-world evidence indicates a broad educational base, but literacy and expenditure are imperfect proxies for the quality of scientific training. The game's education mechanisms are more specific: secondary education raises domestic human capital, while tertiary education can also affect international talent when visitors are present. Because both improve research success, their value depends on the size and timing of the improvement relative to its cost and the number of research attempts remaining. Russia should compare the next affordable secondary and tertiary upgrade, rather than automatically buying either. [R1; G1; AI-2]"

"For Round 1, my proposed education envelope is 20% of newly issued spending points, provisionally divided equally between secondary and tertiary education. This is a planning allocation, not a claim about legal unit purchases. If the cheapest effective upgrade exceeds the envelope, choose the better affordable upgrade, or defer and retain the points; do not sacrifice every applied-research attempt to satisfy percentages. I would favor secondary education when it produces a larger domestic human-capital gain per point, and tertiary education when its research benefit or verified talent-flow effect is stronger. The United States' starting university advantage, China's population scaling, and France's inexpensive secondary upgrades show why a common education budget across countries would be inappropriate. [G1; AI-2]"

"This strategy is preferable to either neglecting education or maximizing it regardless of horizon. An early improvement can benefit several later research attempts, while a late investment may never repay its cost in a short game. Before increasing the education share, I will calculate the expected additional technology points generated over the remaining plausible rounds and compare that gain with direct research and retained wealth. If no productive payback is visible, I recommend maintaining current capacity and shifting the allocation toward research or savings. I will coordinate with Gloria before using university investment to justify openness, because visitor eligibility and talent retention must follow the actual game formula. [G1-G2; AI-2]"

AI generated review draft. Quoted italic text requires independent student review. Citations, exact prompt, and factual critique follow in the appendices. No team consensus is asserted.

Technology strategy recommendation for Steve

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"I recommend a focused applied-research strategy supported by only the basic research and human capital that make it productive. Russia's +7 starting tacit knowledge is a game advantage, not a guarantee of research success or automatic access to advanced categories. WIPO's indicators show a gap between technical capability and wider innovation performance. European Commission guidance updated in September 2026 also documents restrictions on external access to dual-use and advanced technology; the game does not explicitly implement these restrictions. We should use our legacy capabilities to generate affordable output, while avoiding an attempt to match the United States' budget or China's scale across all research paths. [R5, R9; G1; AI-3]"

"My Round 1 proposal assigns 45% of newly issued spending points to L applied research and 15% to basic research, leaving the rest for education and a reserve. L is the only initially available applied category in the supplied rules. Basic research has two distinct benefits: it helps meet advanced-path requirements and gives the investing country tacit knowledge, whereas foreign basic research contributes only the stated half-strength public spillover. I therefore recommend using foreign spillovers where available while retaining domestic basic research when its marginal unlock or tacit-knowledge value warrants the cost. If the next basic-research purchase cannot change a useful threshold or research probability, redirect it toward L research or saving. [G1; AI-3]"

"Once M or H becomes legally available, I will compare expected technology points and their final-score contribution per spending point, accounting for failure risk and the remaining horizon. Higher rewards do not establish a better expected return. We should move into M when it offers a better marginal payoff, and consider a bounded H position only when its prerequisites, odds, and potential contribution justify the risk. A short game favors earlier realizable returns; a longer game can reward investment in capabilities. Gilli and Gilli's argument about tacit and organizational knowledge cautions against treating foreign designs as a substitute for domestic capability. The game's espionage mechanism is more generous than that real-world process, so it should be evaluated on its own published terms. [C1; G1-G2; AI-3]"

AI generated review draft. Quoted italic text requires independent student review. Citations, exact prompt, and factual critique follow in the appendices. No team consensus is asserted.

Intelligence and immigration recommendation for Gloria

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"I recommend retaining closed immigration in Round 1 unless the actual talent-flow formula demonstrates a better alternative. Closure provides the stated defensive bonus, but I do not assume it prevents all talent outflow. Opening costs zero dice to change, yet affects human capital and espionage exposure. Russia's wealth and university quality may make openness less attractive than for the United States or France. The migration statistic cannot independently verify scientist flows. Separately, the UK government's July 2026 GRU profile describes Russian military intelligence cyber capabilities; this attributed assessment supports the relevance of intelligence, not a numerical espionage success rate. My recommendation rests on the game's actual flows and payoffs. [R1, R10; G1; AI-4]"

"For intelligence collection, I recommend no mandatory Round 1 espionage spending. We first need the roll thresholds, country modifiers, category-level rewards, and legal targeting rules. If an affordable attempt has a better expected score contribution than a marginal research purchase or saving, the United States is a plausible initial candidate because its large budget may support valuable programs; this is a hypothesis, not knowledge of its secret allocations. France may be attractive because it begins open, and China remains relevant if its accumulated research becomes valuable. We may target at most two programs under the stated constraint, subject to the instructor's interpretation of targets. Successful observation reveals spending history; only a critical success copies the specified historical category earnings. Ordinary success must not be counted as stolen technology points. [G1; AI-4]"

"My operating rule is to purchase espionage only against a justified target and to reassess after any exposure or policy change. Critical failure reveals our behavior; its modeled penalty and any effect on other teams' future choices must enter the comparison. I will distinguish publicly visible basic research from hidden applied research, label uncertain estimates, and avoid asserting that another team is leading without evidence. Immigration policy will be reviewed with Tiger after education decisions, especially if stronger universities change the expected talent balance. The combined portfolio should protect and augment domestic capability; it should not displace research with a sequence of speculative intelligence gambles. [G1; AI-4]"

AI generated review draft. Quoted italic text requires independent student review. Citations, exact prompt, and factual critique follow in the appendices. No team consensus is asserted.

Executive recommendation for Steve with team review

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"I propose adopting the majority-and-tiebreaker rule stated in the assessment: all three members advise and vote; two votes adopt a decision, while I resolve a remaining deadlock or expired decision period after hearing the alternatives. My recommended overall strategy is to convert Russia's existing tacit knowledge into feasible applied-research returns, invest selectively in the capabilities needed for later opportunities, and retain wealth. Round 1 planning shares are 45% L applied research, 15% basic research, 10% secondary education, 10% tertiary education, and 20% saved, with no compulsory espionage purchase and provisionally closed immigration. These shares must be converted into legal purchases using the real budget counter; they are not an executable order until that check is complete. [G1; AI-5]"

"I prefer this strategy because it addresses all three stated scoring considerations: technology points, retained wealth, and improvement relative to the starting position. It also gives us an opening that can produce returns if the game ends sooner than expected. We will change the allocation when a documented threshold, payoff, talent flow, or observable rival action changes the marginal return. Savings will be compared with research in score units once the wealth-conversion and final-score formulas are known. We will not assume that merely holding unspent spending points increases normalized W or the next income payment; that conversion must be confirmed. The largest early uncertainty is the payoff from education and basic research relative to direct production, rather than a presumption that radical research must eventually win. [G1-G2; AI-5]"

"This is a proposed synthesis prepared before the team meeting; it does not record a completed internal debate. Tiger, Gloria, and I have not yet recorded whether we agree with every recommendation, and no consensus is asserted. At the meeting, we will review each portfolio, record the votes and objections, and revise this paragraph to state the actual outcome. If a recommendation is rejected, the final brief will explain the alternative chosen and the decision rule used. Each member will verify and revise their own portfolio, and we will share the work of checking this executive section and taking gameplay notes so the combined roles do not concentrate all supporting work on one person. [S1, p. 4; A1; AI-5]"

AI generated review draft. Quoted italic text requires independent student review. Citations, exact prompt, and factual critique follow in the appendices. No team consensus is asserted.

References and source provenance

All web sources accessed November 10, 2026. Fixed editions are stated explicitly; these are not claims to provide the newest 2026 statistical release. The evidence register explains source limitations and gives game-page references.

S1. Canfil, Justin K. *84-375 / 84-675 Geopolitics of Innovation*, syllabus v1.8, August 31, 2026, designated most-updated PDF. Assignment pp. 2-4; AI policy p. 6; schedule pp. 16-18.

G1. Canfil, Justin K. *SIMULATION RULES*, supplied two-page PDF, pp. 1-2.

G2. Canfil, Justin K. *SIMULATION PREP*, supplied two-page PDF, pp. 1-2.

A1. Canfil, Justin K. *Final syllabus update*, announcement preserved in semester-readings/sources/schedule-final-update-received-2026-11-04.md; source timestamp not supplied. Role allocation separately follows the November 4 screenshot supplied by the user.

R1. CIA, *The World Factbook*, Russia, archived data reproduced by theworldfactbook.org, independent mirror, not an official CIA website. Fields: education expenditure, literacy, school life expectancy, GDP at official exchange rate, real GDP PPP, net migration, space program.

R2. CIA, *The World Factbook*, United States, archived data via independent mirror, education expenditure.

R3. CIA, *The World Factbook*, China, archived data via independent mirror, education expenditure.

R4. CIA, *The World Factbook*, France, archived data via independent mirror, education expenditure and real GDP PPP.

R5. WIPO. *Russian Federation ranking in the Global Innovation Index 2025*, pp. 1, 5. Use the title inside the PDF; the search-result title incorrectly says 2024.

R6. WIPO. *United States of America ranking in the Global Innovation Index 2025*, p. 1.

R7. WIPO. *China ranking in the Global Innovation Index 2025*, p. 1.

R8. WIPO. *France ranking in the Global Innovation Index 2025*, p. 1.

R9. European Commission. *Export-related restrictions for dual-use goods and advanced technologies*, updated September 25, 2026.

R10. UK government. *Profile GRU cyber and hybrid threat operations*, updated July 13, 2026; sections on established structures and Unit 26165. This is an attributed UK government assessment.

C1. Gilli, Andrea, and Mauro Gilli. 2019. "Why China Has Not Caught Up Yet: Military-Technological Superiority and the Limits of Imitation, Reverse Engineering, and Cyber Espionage." *International Security* 43(3): 141-189, particularly pp. 141-143. Local published-text extract: semester-readings/sources/gilli-gilli-2019.txt.

AI disclosure and prompt record

Prepared November 10, 2026. This appendix implements the designated syllabus's p. 6 disclosure requirement. It is part of the first draft, not proof of completed student review.

Tool and scope

OpenAI ChatGPT generated and organized this first-draft preparation packet on November 10, 2026. A precise model version was not independently recorded. It read the supplied syllabus, both files initially in the simulation folder, the role-assignment screenshot, and the saved professor update; it retrieved cited external sources and inspected relevant local published reading extracts. It drafted recommendations and supporting work materials. It did not attend a team meeting, obtain teammate approval, carry out the simulation, create a native shared Google doc, or send an email.

Exact user prompt

The request below is reproduced verbatim, including spelling and punctuation:

most-updated-_Syllabus_ 84-375 Geopolitics of Innovation (2026) (1).pdf

this is the most updated syllabus, take that as the benchmark

@simulation this is the folder that you will work on

there are already files in simualtion folder, make sure you read all of them

So I think the professor wants us to, like, write something before the simulation. And I want to write it all. Like I want to basically have a first draft that is comprehensive for my team, like for Steve, Gloria, and Tiger, you know. And, like, I think the role assignments are already in the screenshot, and, you know, just basically finish all the preparations for me, you know, that is before the simulation. Yeah. I think later, like my team is gonna have a meeting to discuss, go through, like my first draft, you know. Just make sure you have everything ready. Yeah. And make sure you commit and push.

The same request included a screenshot of the professor's November 4 simulation team-assignment announcement. Its relevant row assigns Russia to Steve, Gloria, and Tiger; education to Tiger, technology strategy and executive to Steve, and intelligence/immigration to Gloria. The screenshot also recommends coordinating a holistic spending strategy before Day 1 and selecting a decision rule.

No separate human revision prompt was supplied during the initial drafting turn. Automated search strings and file-reading commands were research operations, not additional student-authored prompts. Retain this conversation if a complete interaction transcript is requested.

Instance-specific citations

AI-1 - Brief, one-page country assessment - Synthesized cited comparative evidence and game-country tradeoffs; drafted five paragraphs and a proposed voting rule.

AI-2 - Brief, education recommendation - Drafted Tiger's three paragraphs, provisional allocation, and marginal-payback reasoning.

AI-3 - Brief, technology recommendation - Drafted Steve's three paragraphs, BR/AR priorities, and conditional research transitions.

AI-4 - Brief, intelligence/immigration recommendation - Drafted Gloria's three paragraphs, conditional IM posture, and in-game SP evaluation.

AI-5 - Brief, executive recommendation - Drafted three paragraphs integrating portfolios, proposed process, and truthful pending-consensus status.

AI-6 - Research and evidence register - Organized source provenance, statistical limitations, comparative game table, and rules ambiguities.

AI-7 - Game playbook - Proposed budget shares, arithmetic example, decision criteria, contingency rules, and instructor questions.

AI-8 - Meeting agenda and checklist - Drafted team-review process, alternative allocation table, and proposed equitable work division.

AI-9 - Gameplay record - Created blank evidence and report-planning templates. No outcomes were generated.

AI-10 - README, references, and this appendix - Organized requirements, source citations, date conflicts, navigation, and disclosure language.

The assignment-facing substantive paragraphs are enclosed in quotation marks and italicized. This is deliberate compliance with the policy for copied substantive AI output. Internal planning tables are also generated material: if copied into an assessed submission, cite the relevant instance and apply the required distinguishing format there as well. Independently rewriting a section does not erase the fact that AI contributed to drafting; describe its role accurately in the final disclosure.

Factual critique of the draft

1. **Unknown mechanics limit optimization.** The 45/15/10/10/20 plan is a proposed starting mix. The PDFs omit prices, starting W, score weights, success odds, migration calculations, and exact threshold timing. No optimality proof or executable opening order is claimed.

2. **The provided instructions contain conflicts.** The designated syllabus and saved announcements differ on dates and research sources. PREP and RULES differ on round count, and RULES contains ambiguous wording about unlocks, categories, and critical espionage. The packet records the conflicts instead of fabricating a resolution.

3. **Sources have different status.** WIPO country PDFs are primary publications. The Factbook country pages used are independent reproductions after CIA retirement, not official current pages. The UK intelligence profile is an attributed government assessment. The EU publication establishes technology restrictions, not their precise innovation effects.

4. **Historical observations are not game variables.** GDP at official exchange rates differs from PPP GDP. Neither directly generates the instructor's normalized W. Education-spending observations have different years and cannot establish education quality. GII rankings do not provide dice probabilities. Fixed 2025 GII comparisons are not labeled latest 2026 rankings.

5. **Migration evidence is limited.** Positive aggregate net migration can coexist with skilled emigration. The packet does not claim to measure scientist migration or describe a complete current visa regime. The binary IM variable is a game abstraction.

6. **Recommendations are conditional.** Closure is not assumed to prevent talent loss; espionage is not assumed to succeed; a higher-risk category is not assumed to have a higher expected return. Real-world research supports context, not invented bonus mechanics.

7. **Student participation remains to happen.** The named portfolio owners have not yet independently verified or rewritten these drafts. Agreement, dissent, vote outcomes, submitted orders, and simulation results remain blank or explicitly unrecorded.

Student review and later revision record

Complete these entries after the team's actual work. Do not replace blanks with generated claims of review.

Tiger

Gloria

Steve

For every later AI-assisted revision, append the actual tool/date, exact prompt, affected section, what was retained, and any corrections made after checking evidence. Do not claim the first prompt covers later interactions that have not been logged.